Accident Risk Prediction and Hotspot Analysis

Introduction

- Accidents depend on traffic, weather, and time patterns.
- Manual prediction is difficult.
- Aim: Predict risk level & identify hotspot conditions.

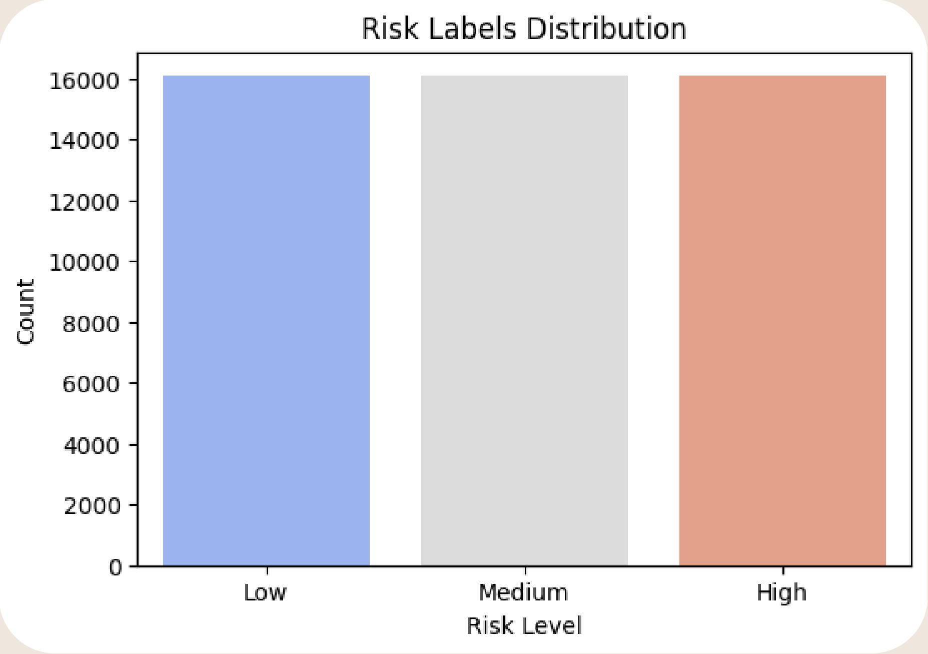
Dataset & Preprocessing

- Dataset: Metro Interstate Traffic Volume (UCI/Kaggle).
- Features: traffic, weather, clouds, rain, snow, hour, day.

	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
0	NaN	288.28	0.0	0.0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
1	NaN	289.36	0.0	0.0	75	Clouds	broken clouds	2012-10-02 10:00:00	4516
2	NaN	289.58	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 11:00:00	4767
3	NaN	290.13	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 12:00:00	5026
4	NaN	291.14	0.0	0.0	75	Clouds	broken clouds	2012-10-02 13:00:00	4918

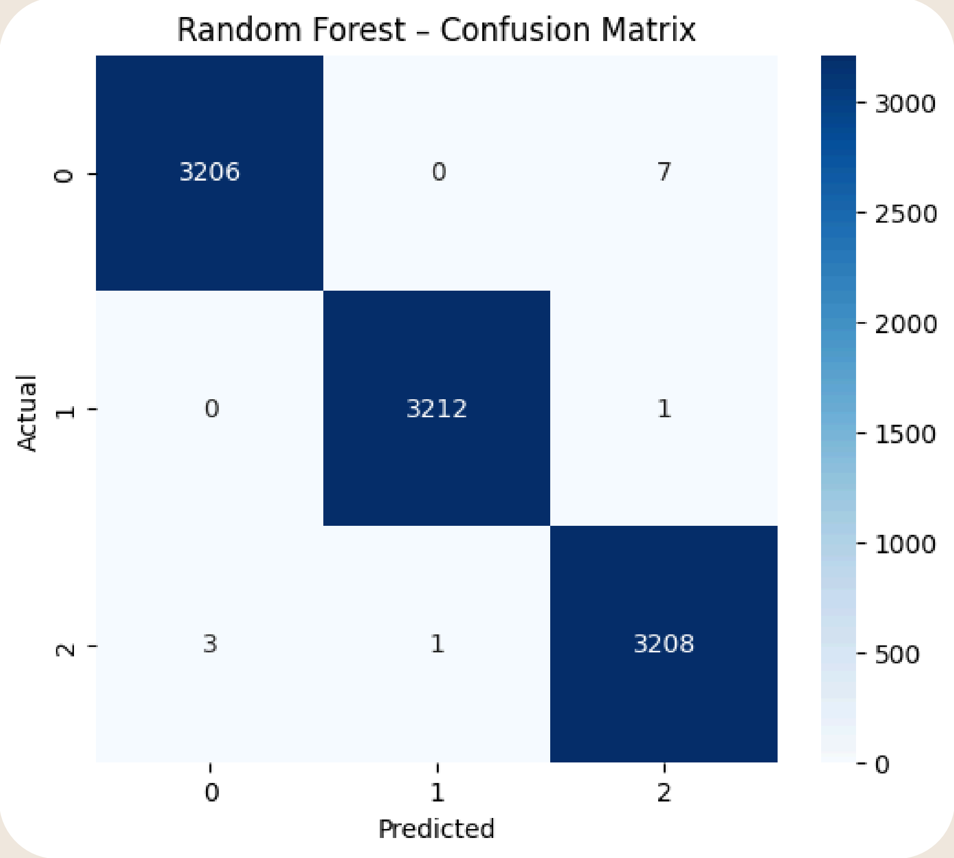
PCA Risk Scoring

- PCA used to combine multiple features into one risk score.
- Avoids manual weighting → more objective.
- Risk labels: Low / Medium / High.



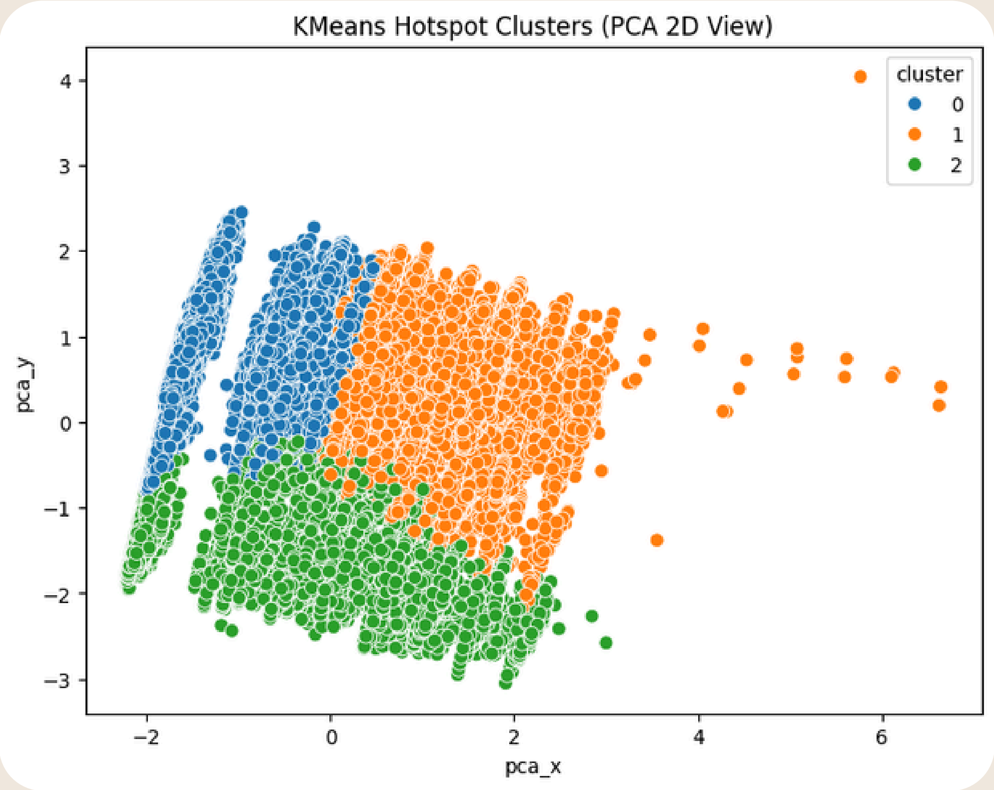
ML Models & Results

- Models trained: Logistic Regression, Random Forest, SVM.
- Train/Test split with scaling.
- Random Forest performed best overall.



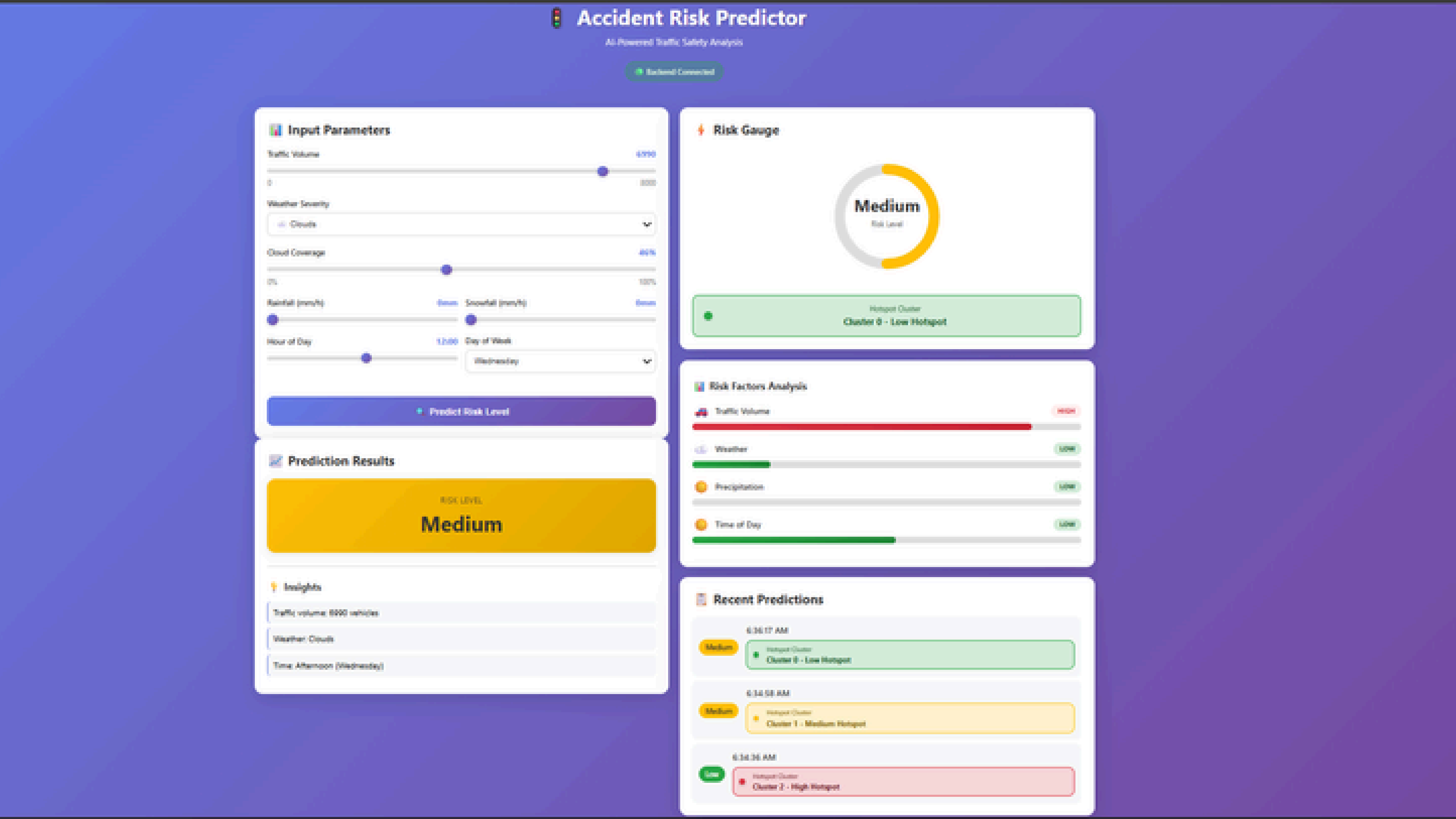
Hotspot Clustering (KMeans)

- KMeans (k=3) used to find natural condition clusters.
- Cluster meanings: Low, Medium, High hotspot.
- Shows general pattern group, not current moment risk.



System Architecture

- React frontend → Flask backend → ML model → prediction.
- User inputs conditions → system outputs risk + hotspot.
- Real-time, fast, and explainable.



Accident Risk Predictor
AI-Powered Traffic Safety Analysis
Input Parameters
Traffic Volume
Weather Severity
Clouds
Cloud Coverage
Rainfall (mm/h)
Snowfall (mm/h)
Hour of Day
Day of Week
Wednesday
Predict Risk Level
Prediction Results
RISK LEVEL
Medium
Insights
Weather: Clouds
Time: Afternoon (Wednesday)
Risk Gauge
Medium
Risk Level
Hotspot Cluster
Cluster 0 - Low Hotspot
Risk Factors Analysis
Traffic Volume
HIGH
Weather
LOW
Precipitation
LOW
Time of Day
LOW
Recent Predictions
Medium
Cluster 0 - Low Hotspot
Medium
Cluster 1 - Medium Hotspot
Low
Cluster 2 - High Hotspot

Conclusion & Future Work

Conclusion:

- Working ML system for risk prediction.
- PCA scoring, Random Forest best accuracy.
- Clustering revealed risk patterns.

Future Work:

- Add real-time weather & traffic APIs
- Show hotspots on Google Maps
- Add mobile interface & alert system

Thank you